Question Paper Code: 526427

B.E./B.Tech. DEGREE EXAMINATIONS, APRIL/MAY 2026

Fourth Semester
Computer Science and Engineering
CS23413 – THEORY OF COMPUTATION
(Regulations 2023)

Time: Three Hours **Maximum:** 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Why is Automata Theory important?

2. Show that ε-NFA and NFA are equivalent.

3. What is Pumping Lemma?

4. What is closure property?

5. What are the types of grammar in Chomsky Hierarchy?

6. What are the components of PDA?

7. What is the difference between null production and unit production?

8. List out the components of a Turing Machine.

9. What is Modified PCP (MPCP)?

10. State the difference between recursive and RE language.

PART B – (5 × 13 = 65 Marks)

11(a). Define DFA and NFA. Prove that NFA is equivalent to DFA using subset construction with an example.
Or
11(b). Explain DFA minimization using partitioning method and illustrate with an example.

12(a). Explain conversion of Regular Expression to Finite Automaton using Thompson's construction.
Or
12(b). Simplify the given regular expression: (a|b)(a|b)*.

13(a). Define CFG and explain derivations and parse trees with an example.
Or
13(b). Construct PDA for language $L = \{a^n b^n \mid n \geq 0\}$.

14(a). Explain Pumping Lemma for CFL and prove that $L = \{a^n b^n c^n\}$ is not a CFL.
Or
14(b). Design a Turing Machine to recognize strings of the form $a^n b^n$.

15(a). Explain Post Correspondence Problem (PCP) and prove that it is undecidable.
Or
15(b). Prove that 3-CNF-SAT is NP-complete.

PART C – (1 × 15 = 15 Marks)

16(a). Prove closure properties of regular languages under union, concatenation, and Kleene closure with examples.
Or
16(b). Define a Turing Machine. Design a TM that accepts the set of palindromes over {0,1}.